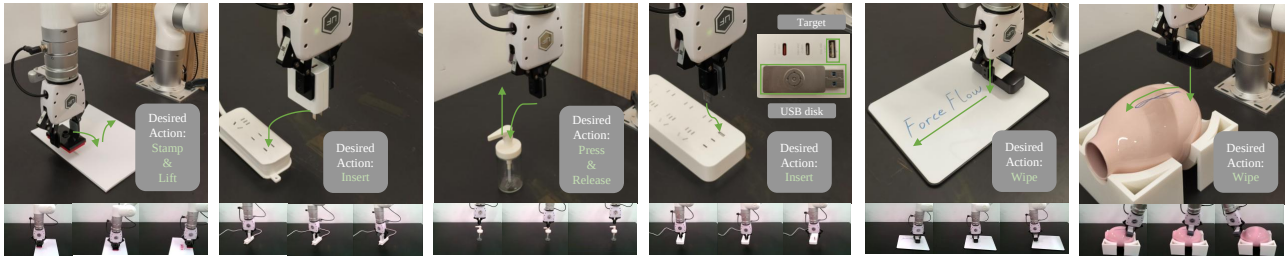

(a) Stamp (b) Plug (c) Press (d) Insert (e) Clean WB (f) Clean Vase

Figure 3. **Contact-Rich Task Suite.** We evaluate ForceFlow on six diverse manipulation tasks categorized into: (a)-(d) **Short-Horizon Contact Tasks** requiring precise contact establishment; and (e)-(f) **Continuous Contact Tasks** demanding consistent normal force tracking along trajectories.

Table 1. **Quantitative Comparison of Task Success Rates (SR).** We report the success rate (%) over 20 trials for each task. **Bold** indicates the best performance.

Method	Stamp	Plug	Press	Insert	Clean WB	Clean Vase	Avg.
$\pi_{0.5}$	0%	60%	30%	45%	10%	0%	24.17%
ACT	0%	30%	5%	0%	15%	0%	8.33%
Diffusion Policy	0%	40%	20%	50%	75%	0%	30.83%
ForceVLA	20%	70%	65%	15%	100%	0%	45%
ForceFlow (w/o Force)	20%	75%	0%	40%	100%	30%	44.17%
ForceFlow (Ours)	**85%**	**90%**	**90%**	**60%**	**100%**	**65%**	**81.67%**

4.1. Experimental Setup

Task Suite (Figure 3). We select six tasks categorized into: (1) short-horizon contact tasks that require precise modulation to establish contact (Stamping, Plug/USB Insertion, Press Button); and (2) continuous contact tasks which demand consistent normal force (Clean Whiteboard, Clean Vase). Detailed task settings are provided in Appendix A.

4.2. RQ1: Effectiveness in Contact-Rich Tasks

We compare ForceFlow against $\pi_{0.5}$ (Black et al., 2025), ACT (Zhao et al., 2023), Diffusion Policy (Chi et al., 2023), ForceVLA (Yu et al., 2025), and the ablation baseline ForceFlow (w/o Force). As presented in Table 1, ForceFlow achieves an average success rate of **81.67%**, significantly outperforming the best baseline (ForceVLA, 45%). We analyze this performance gain through three functional